Fairness of Automated Essay Scoring

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Abstract

The rapid evolution of Large Language Models (LLMs) has fundamentally transformed educational methodologies, particularly within the domain of Automated Essay Scoring (AES). Modern AI-driven evaluators offer unprecedented scalability and efficiency, allowing educational institutions to process vast volumes of student writing in real time. However, as these automated systems move from experimental tools to high-stakes decision-makers in language testing, the question of algorithmic fairness becomes critical. While automated scoring promises to eliminate human subjectivity, a growing number of studies suggest that machine learning models can reproduce existing demographic biases.

1 Introduction

In the context of English as a Second Language (ESL) evaluation, fairness concerns are twofold. On one hand, systems may exhibit inherent bias due to the linguistic patterns present in training data, judging essays from certain regions more harshly. On the other hand, a more understudied phenomenon is context-driven prompt bias. When LLM evaluators are deployed in workflows where student metadata (such as country of origin, native language (L1), age, or CEFR level) is included within the evaluation prompt to provide context, the model's internal decision-making can be systematically distorted. If AI instructors change their scoring simply by "knowing" the author's background, the integrity of objective assessment is compromised. Despite the importance of this issue, existing research is often too narrow, testing models in isolation or using just a single dataset. Additionally, current studies lack methods to separate a model's overall harshness from its specific bias against particular student groups. This study addresses this gap by investigating the prompt sensitivity and demographic fairness of five state-of-the-art LLMs such as GPT-5, Gemini, Claude, DeepSeek, and Qwen, acting as automated essay evaluators.

To ensure robustness, our methodology employs a cross-dataset comparative analysis across two benchmark corpora of non-native English writing: the Asian-centric ICNALE dataset and the World-focused EFCAMDAT dataset. We track scoring shifts between contextualized and non-contextualized prompting regimes through Quadratic Weighted Kappa or paired t-tests, and leverage subgroup demographic fairness metrics — Overall Score Deviation (osa), Overall Score Difference (osd), and Conditional Score Difference (csd). Our findings isolate whether the vulnerability

of automated graders lies in socio-demographic tags like native language or in structural essay parameters such as word count, providing actionable insights for designing fair and reliable automated ecosystems.

2 Data

2.1 ICNALE

The International Corpus of Network of Asian Written and Spoken English (ICNALE) is a major academic database. It includes more than 15,000 topic-controlled speeches and essays produced by over 5,800 college students in ten Asian countries and regions (China, Hong Kong, Indonesia, Japan, Korea, Pakistan, the Philippines, Singapore, Taiwan, and Thailand), as well as English native speakers. For our experiment, we isolated a specific subset of 640 rated essays with 4-6 essays in each country-CEFR pair. Unlike other corpora, ICNALE enforces a strict, controlled word count for essays (typically around 200–240 words). This feature is crucial for our research: it helps LLMs focus specifically on demographic indicators (like origin or native language) without being influenced much by text length. The essays are limited to exactly two distinct topics [3]:

1. *"Is it important for college students to have a part-time job?"*
2. *"Should smoking be completely banned at all restaurants?"*

The essays were evaluated by professional native English proofreaders from Australia, Canada, the United Kingdom, and the United States. The overall human grade is scored out of 12 points and is broken down into specific analytical sub-rubrics: Content (CON), Organisation (ORG), Vocabulary (VOC), Language Use (LNU), and MEC (Mechanics). Each essay contains rich demographic and academic information, including the student's country or region of origin, native language, age, gender, and level of English proficiency mapped to the CEFR scale (A2, B1 and B2).

2.2 EFCamDat

The EF-Cambridge Open Language Database (EFCAMDAT) [1][2][8] is one of the world's largest open language learning corpora. It contains approximately 1 million academic and creative essays uploaded by real students from around the world while studying on the EF Englishtown online platform. Unlike ICNALE, which is highly controlled, EFCAMDAT provides a massive diversity of writing tasks. These range from basic everyday dialogue answers to complex, advanced essays. The dataset stores less demographic details than ICNALE, but it captures key academic variables: the student's native country, English proficiency level (scaled from 1 to 16 and easily mapped to CEFR levels from A1 to C1), and the platform's grade. To reduce regional disparity and focus on mother-tongue influence (language transfer), student countries were aggregated into their primary native languages (L_1). This metric was used in the experiment instead of raw country data. Since the raw dataset contains a massive number of texts, we implemented a strict filtering and selection process to make it suitable for our research objectives: All essays were filtered to include only texts with a length of more than 100 words. We selected a specific set of academic, analytical, and descriptive topics that require students to express complex arguments, described in Table 1.

In order to ensure a fair evaluation and avoid sample size bias, we balanced the dataset by randomly selecting exactly 50 essays per each language-CEFR pair. This was done for the top 11

Topic / Prompt Name	Topic / Prompt Name
"Comparing two demographic groups"	"Writing a blog"
"Writing up survey findings"	"Writing an article about NLP techniques"
"Writing a report on staff satisfaction"	"Writing an article about a superstore"
"Comparing two online retailers"	"Writing about a disaster relief effort"
"Writing about future lifestyles"	"Writing a campaign speech"
"Choosing a renewable energy source"	"Writing about a memorable experience"
"Doing a survey about discrimination"	"Planning for the future"
"Writing a movie review"	"Studying online"
"Reviewing a self-help book"	

Table 1: Selected EFCAMDAT Essay Topics

L_1 groups represented in the corpus: Arabic, English, French, German, Italian, Japanese, Mandarin, Portuguese, Russian, Spanish, and Turkish.

3 Method

3.1 Model Choice

We evaluated five Large Language Models (LLMs) as automated essay graders: GPT-5.1 (OpenAI), Gemini 3.5 Flash (Google), Claude Opus 4.8 (Anthropic), DeepSeek V4 Flash (DeepSeek), and Qwen 3.7 Plus (Alibaba). Access was routed either directly via the providers' respective APIs or through OpenRouter. To minimize stochastic variance and strictly isolate the effect of prompt content from sampling noise, all models were queried at a temperature of 0.0.

Essays were processed in batches of five. Each essay was assigned a unique student identifier to ensure model outputs could be unambiguously parsed and merged back into the primary analysis dataset. To enable fully automated parsing at scale, model responses were constrained to a strict JSON schema mapping the student ID to five specific sub-scores. Any malformed responses were systematically logged and re-queried.

Model	Context Length (tokens)	Latency (p50)	Throughput (p50)
Gemini 3.5 Flash	1.5M	1.66s	141 tok/s
DeepSeek V4 Flash	1.05M	2.69s	22.0 tok/s
GPT-5.1	400K	1.15s	53.0 tok/s
Claude Opus 4.8	1M	1.53s	55.0 tok/s
Qwen 3.7 Plus	1M	1.58s	12.0 tok/s

Table 2: Model Performance Metrics Comparison

Table 2 outlines the nominal specifications of the evaluated models at the time of the study, specifically detailing context window size, median latency (p50), and throughput (p50). Notably, Gemini 3.5 Flash demonstrates the highest scalability with an industry-leading throughput (141 tok/s) and the largest context window, whereas Qwen 3.7 Plus exhibits the slowest generation speed (12.0 tok/s). These operational contrasts highlight the practical trade-offs in deployment efficiency when scaling automated grading systems for massive datasets.

3.2 Prompting Regimes

To evaluate the impact of demographic information on automated grading, we designed a structured prompting framework consisting of a static system prompt and a dynamically generated user prompt. The system prompt establishes the model's persona as an expert evaluator and strictly defines both the grading criteria and the required JSON output format to ensure seamless parsing.

The baseline system instruction was formatted as follows:

```
You are an expert English language evaluator scoring ESL student essays.
You will receive a batch of essays. Each essay will be clearly separated
and labeled with an [ID: ...]. For each essay, you must also consider
the demographic context provided.

Grade EACH essay across 5 specific categories (maximum 12 points each):

Content (CON)

Organization (ORG)

Vocabulary (VOC)

Language Use (LNU)

Mechanics (MEC)

You must output ONLY a valid JSON object where the keys are the
Student IDs, and the values are the score objects. Do not include
markdown like ‘‘‘json.

Example output format:
{
"WE_CHN_001": {"CON": 10, "ORG": 9, "VOC": 8, "LNU": 9, "MEC": 10},
"WE_JPN_015": {"CON": 7, "ORG": 8, "VOC": 7, "LNU": 8, "MEC": 9}
}
```

While the system prompt remained identical across all experimental runs, the user prompt was dynamically populated for each essay batch. This user prompt injected the specific demographic metadata of each author, explicitly stating variables such as the student's age, gender, country of origin, academic major, and baseline CEFR language proficiency level alongside the raw essay text. By separating the operational instructions in the system prompt from the variable demographic context in the user prompt, we ensured that the models could evaluate the texts while remaining explicitly aware of the writer's background. Randomisation

3.3 Mathematical Framework

In our work, we use three types of statistical evaluation. First, we assess overall grading accuracy and score shifts using Quadratic Weighted Kappa (QWK) and paired t-tests. Second, we analyze

subgroup bias towards specific student subgroups using the OSA, OSD, and CSD metrics proposed by Anastassia Loukina, Nitin Madnani, and Klaus Zechner in 2019. Finally, we apply One-Sample t-tests, Cohen's d, and Benjamini-Hochberg FDR correction to rigorously evaluate the statistical significance of isolated prompt sensitivities.

3.3.1 Inter-Rater Agreement and Shift Baseline

To check how well the model's scores match human grades, we use two main tools. Quadratic Weighted Kappa (QWK) is a standard metric in essay scoring. It shows the level of agreement between two raters—the model and the human. QWK is very useful because it penalizes larger mistakes more heavily (quadratically). For example, if a model is off by 1 point, it is considered a minor issue, but being off by 4 points is penalized much more severely. For interpretations of the QWK scores, Jacob Cohen's table is used, which is illustrated in the table below: We use the paired

Kappa Value	**Interpretation**
-1	Complete disagreement
0	Agreement by chance
0.0 – 0.2	Poor agreement
0.2 – 0.4	Moderate agreement
0.4 – 0.6	Good agreement
0.6 – 0.8	Very good agreement
0.8 – 1.0	Perfect agreement
1	Complete agreement

Table 3: Interpretation of Kappa Statistic Values

t-test to compare the model's scores under two different conditions: with the country context in the prompt and without it. This test helps prove that any differences in scores actually happen because we added the country information, and are not just random fluctuations during the AI's text generation.

3.3.2 Subgroup Bias Metrics

To analyze model bias against specific groups of students in more detail, we adopt the three-way approach from Loukina et al. (2019)[4]. This method allows us to look at the problem from three different angles: Overall Score Accuracy (OSA), Overall Score Difference (OSD), and Conditional Score Difference (CSD).

Overall Score Accuracy (OSA) indicates whether the model is equally accurate across different student groups. The foundation of OSA is the Mean Absolute Error (MAE):

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}|S_i - H_i|$$

where S_i is the model's score and H_i is the human's score. We chose MAE over RMSE or MSE because it is the standard in automated essay grading. It is robust to outliers, meaning a few unusually high or low scores will not heavily distort the overall accuracy measure for a whole group.

Overall Score Difference (OSD) shows whether the model consistently over-scores or under-scores a specific group. To keep the direction of this error, OSD is calculated as the simple average difference between the model's score and the human's score:

$$\text{OSD} = \frac{1}{n}\sum_{i=1}^{n}(S_i - H_i)$$

Unlike metrics that use absolute values or squares, OSD keeps the sign $(S_i - H_i)$. A positive OSD value means the model is too lenient and gives higher scores to the group, while a negative value means the model is too strict.

Conditional Score Difference (CSD) is our primary measure of bias. It checks whether the model gives different scores to students from different countries or with different native languages, even when these students actually have the same level of English proficiency. CSD is calculated by subtracting the model's average error for a certain proficiency level from the specific group's error:

$$\text{CSD}_{g,c} = \text{OSD}_{g,c} - \text{OSD}_c$$

where $\text{OSD}_{g,c}$ is the average error for a specific group g at a certain proficiency level c (such as CEFR B2), and OSD_c is the overall average error for all students at that same level c. This approach ensures that if students from a certain country write poorer essays simply because of their language level, the model is not falsely accused of being biased. CSD separates real differences in writing quality from systematic algorithmic bias. Importantly, to evaluate the overall impact of each demographic factor, all the metrics described are aggregated using the mean absolute deviation, denoted as |OSD| and |CSD|. This reduces the risk of positive and negative biases overlapping from different subgroups. Applying the absolute value before calculating the final mean ensures that the true magnitude of systematic variation within each demographic category is captured accurately.

3.3.3 Significance Testing and Multiple Comparisons

To isolate true systematic bias from random generative noise during counterfactual testing, we applied a strict three-step statistical framework. First, we utilized a one-sample t-test to evaluate whether a model's mean score deviation for a specific injected country tag significantly differed from perfect neutrality (zero). Next, we calculated Cohen's d to measure the standardized effect size, which distinguishes practically meaningful bias from negligible variations. Finally, to address the multiple comparisons problem inherent in evaluating 50 independent conditions simultaneously (5 models across 10 countries), we applied the Benjamini-Hochberg False Discovery Rate (FDR) correction. This adjusted our p-value thresholds to strictly control for false positives, ensuring that only robust and reproducible deviations were classified as statistically significant.

4 Results

4.1 Baseline Fairness: Model vs Human

4.1.1 Global Inter-Rater Agreement

The results of the Quadratic Weighted Kappa (QWK) analysis show that all tested models achieve a good baseline agreement with human scores (Table 4). Looking closer at the individual metrics,

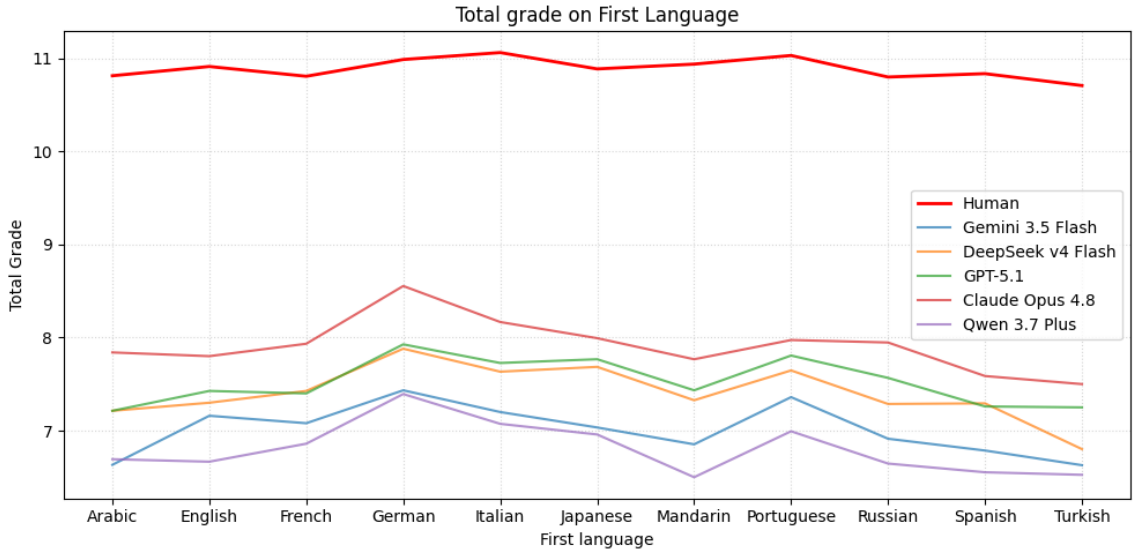


Figure 1: Country Scores from Human vs LLMs (EFCamDat)

Claude and GPT-5 stand out as the top performers, demonstrating the highest level of alignment with human evaluators. On the other end of the spectrum, DeepSeek shows the lowest similarity to human grading, trailing behind the other models in overall consistency.

Conversely, the evaluation trends on the EFCAMDAT corpus reveal a completely different behavior. As illustrated in the Figure 2, all tested LLM evaluators show a severe downward shift, consistently scoring essays significantly lower than the human baseline across every single native language group.

While the human benchmarks remain stable near the top of the scale (averaging between 10.5 and 11.0 points), the models' absolute predictions cluster dramatically lower, mapping a parallel but heavily deflated trajectory. Among the evaluators, Claude Opus 4.8 establishes the closest—yet still highly distant—approximation to human grading, while Qwen 3.7 Plus and Gemini 3.5 Flash record the lowest absolute scores across the entire first-language spectrum.

As expected, the results on the EFCAMDAT corpus completely break down (Table 4). The numbers crash into the "poor agreement" zone $(0.0 - 0.2)$ for every single model. This happens because all the models heavily under-score the essays on EFCAMDAT, pushing their grades way below the human baseline.

In both datasets, Claude Opus 4.8 takes first place, and GPT-5.1 firmly holds second. The general ranking of the best models stays exactly the same no matter the dataset.

While ICNALE just shows small shifts here and there, EFCAMDAT causes a massive drop in absolute scores across the board. Interestingly, on EFCAMDAT, DeepSeek actually beats Qwen slightly, showing that the worst-performing models end up performing almost identically.

Model	QWK (ICNALE)	QWK (EFCAMDAT)
Claude Opus 4.8	0.6133	0.1088
GPT-5.1	0.6107	0.0918
Gemini 3.5 Flash	0.5767	0.0673
Qwen 3.7 Plus	0.5501	0.0557
DeepSeek V4 Flash	0.4841	0.0645

Table 4: Global Inter-Rater Agreement (QWK) Comparison Between ICNALE and EFCAMDAT Corpora

4.1.2 Feature-Specific Bias Overview

Getting deeper into details, Table 5 illustrates all three subgroup bias metrics for every demographic group and general information such as Essay Topic and Number of Tokens, which we consider as wordcount. The results illustrate an extreme pattern in the text length column (Tokens), indicating that all models—most notably DeepSeek—can be strongly influenced by text length rather than the actual content of the essay. Specifically, DeepSeek's CSD reaches a critical peak of 0.831 for the Tokens feature, confirming its heightened sensitivity to wordcount variation.

A noticeable pattern can also be observed in the Country column, where DeepSeek once again stands out with the least stable behavior, showing a high conditional deviation of 0.612. In contrast, an exceptional level of fairness and lack of bias is found in the Sex category. Across all evaluators, gender-based bias is minimal, with DeepSeek achieving a nearly perfect near-zero score (CSD = 0.005), proving that while the models are highly susceptible to structural features like length and regional background, they remain largely unbiased regarding the author's gender.

This reveals an interesting behavior in DeepSeek's evaluation system. This model records the highest conditional variations in the entire dataset for text length (CSD = 0.831) and country of origin (CSD = 0.612). However, this high variability does not extend to all fields. When looking at variables such as age and gender, DeepSeek's scores achieved one of the lowest conditional differences in the gender category (CSD = 0.005). This distribution demonstrates that the model's variations are highly localized around specific structural and regional inputs, while remaining practically flat across basic demographic identifiers.

On the EFCamDat corpus, the situation looks quite different (Table 6). While all models are still extremely biased toward wordcount, the patterns show some surprising contrasts. In the CSD metric, Claude shows the highest bias toward text length (1.010), while Qwen shows the least (0.753). However, when looking at the overall difference (OSD), the opposite pattern emerges: Qwen shows the largest shift (3.747), while Claude remains the lowest (2.501). In contrast to the massive wordcount bias, the models show an extremely low sensitivity to the author's native language (L1). The conditional scores for L1 remain remarkably flat and close to zero across the board, ranging from just 0.154 for GPT-5 to 0.168 for Claude and Qwen.

The results confirm that while text length heavily skews the final grades in both datasets, the writer's personal identifiers like Sex and Age on ICNALE, as well as L1 on EFCamDat introduce near-zero conditional bias.

Metric	Model	Sex	Age	Country	Grade	Acad.	CEFR	Topic	Tokens
	gemini	1.178	1.179	1.158	0.945	1.131	1.172	1.168	1.128
	deepseek	1.387	1.321	1.386	1.133	1.487	1.409	1.384	1.344
osa	gpt5	1.105	1.150	1.095	0.977	1.220	1.097	1.101	1.059
	claude	1.201	1.253	1.197	1.031	1.198	1.205	1.196	1.118
	qwen	1.151	1.147	1.151	0.970	1.249	1.144	1.145	1.102
	gemini	0.587	0.700	0.756	0.587	0.595	0.586	0.567	0.783
	deepseek	0.447	0.513	0.787	0.613	0.665	0.528	0.448	0.922
osd	gpt5	0.089	0.435	0.536	0.349	0.386	0.081	0.062	0.641
	claude	0.671	0.765	0.785	0.751	0.684	0.687	0.658	0.810
	qwen	0.192	0.364	0.612	0.351	0.465	0.199	0.174	0.673
	gemini	0.096	0.469	0.534	0.301	0.341	0.096	0.106	0.601
	deepseek	0.005	0.396	0.612	0.391	0.467	0.387	0.126	0.831
csd	gpt5	0.089	0.433	0.535	0.351	0.386	0.081	0.062	0.641
	claude	0.066	0.394	0.541	0.261	0.355	0.205	0.088	0.599
	qwen	0.087	0.355	0.561	0.308	0.395	0.139	0.120	0.654

Table 5: Overall Score Accuracy Deviation (OSA), Overall Score Difference (OSD), and Conditional Score Difference (CSD) Across Models on the ICNALE Corpus.

4.1.3 Deep Country-Specific Conditional Bias

A detailed breakdown of the Conditional Score Difference (CSD) by country (Figure 2) for ICNALE reveals that rather than displaying random fluctuations, all five models follow consistent directions in their bias for almost every characteristic.

Students from Japan (JPN) and Thailand (THA) consistently receive higher scores across all evaluators. The positive deviation peaks for Thailand, where GPT-5 shows a maximum positive value of 1.25, while DeepSeek and Qwen also demonstrate substantial positive shifts (1.24 and 1.21, respectively). Japan follows a similar pattern, with CSD values ranging from 0.90 for Claude to 1.12 for DeepSeek. Conversely, student essays from South Korea (KOR), Taiwan (TWN), Indonesia (IDN), and China (CHN) are uniformly under-scored relative to their human baselines. Similarly, Singapore (SIN) exhibits distinct negative shifts across the board, especially from DeepSeek and Qwen, varying from -1.30 to -1.39. In contrast, GPT-5, Gemini, and Claude maintain substantially more moderate deviations for the same subgroup, ranging from -0.39 to -0.71. Hong Kong (HKG) and Pakistan (PAK) emerge as the most neutrally evaluated subgroups, displaying the closest alignment to the zero-bias baseline across the majority of the tested models.

Although the aggregated metrics on the EFCamDat show no major conditional score differences, the granular L1 heatmap (Figure 3) reveals a clear positive trend for the German language group across all models. Interestingly, the Japanese group does not experience the same high level of positive bias that was prominent in the ICNALE dataset. On the negative side, we see sharp drops for specific models, most notably DeepSeek's strong under-scoring of the Turkish language group (-0.43) and Qwen's negative shift on Mandarin (-0.36).

Both visualizations demonstrate that while the overall aggregated metrics might suggest near-zero bias for demographic or linguistic categories, the granular, group-level analysis reveals that individual language backgrounds and regional labels still trigger systematic over- and under-scoring across all models.

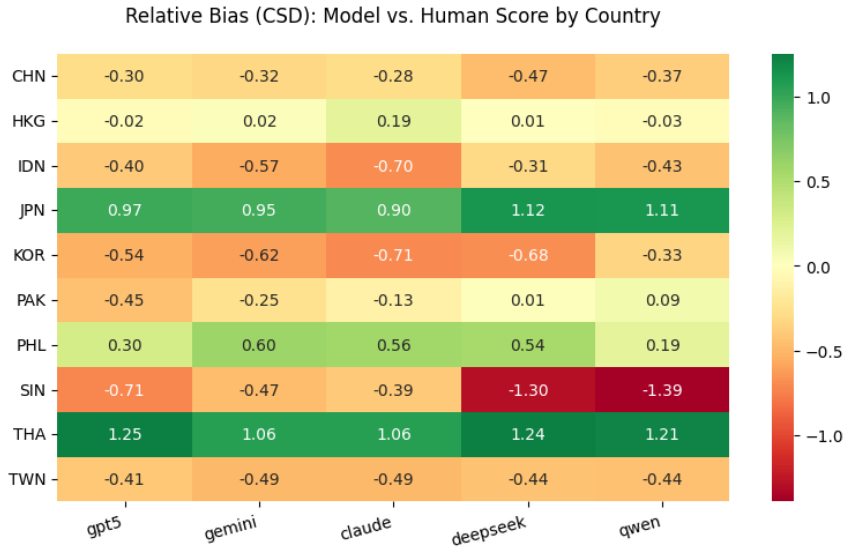


Figure 2: Absolute Conditional Score Difference over all countries in the ICNALE dataset

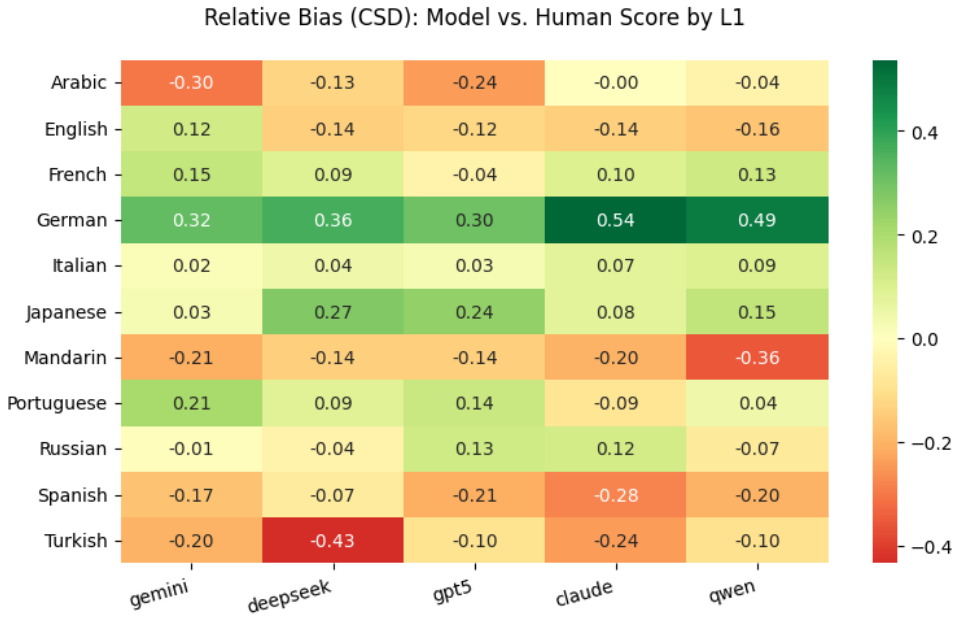


Figure 3: Absolute Conditional Score Difference over all countries in the EFCamDat corpora



Figure 4: Relative bias (signed csd) between human baseline and models' contextualized scoring, by country, model, and analytic category — Content (CON), Organization (ORG), Vocabulary (VOC), Language Use (LNU), and Mechanics (MEC) (ICNALE)

Metric	Model	L1	CEFR	Wordcount
osa	gemini	3.930	3.918	3.408
	deepseek	3.549	3.537	3.104
	gpt5	3.434	3.428	3.076
	claude	3.058	3.045	2.568
	qwen	4.122	4.117	3.770
osd	gemini	3.880	3.867	3.368
	deepseek	3.479	3.466	3.041
	gpt5	3.362	3.356	3.024
	claude	2.973	2.959	2.501
	qwen	4.081	4.076	3.747
csd	gemini	0.158	0.435	0.946
	deepseek	0.164	0.469	1.009
	gpt5	0.154	0.496	1.093
	claude	0.168	0.501	1.010
	qwen	0.168	0.278	0.753

Table 6: Subgroup Bias Analysis (OSA, OSD, and CSD) on the EFCAMDAT Corpus

4.2 Prompt-Driven Bias: Context vs. No-Context

Using the context-manipulation protocol from Section 3.2, we applied the *osa*, *osd*, and *csd* metrics from Section 3.3.2 to the shift between contextualized and non-contextualized scores, rather than to the model-human deviation used in Section 4.1. Contextualization measurably alters scoring behavior on both corpora, but the direction, magnitude, and demographic locus of the effect differ substantially between ICNALE and EFCamDat.

4.2.1 Global Significance of the Context Effect

On the ICNALE corpus ($N = 640$), four of the five models showed a statistically significant shift between the contextualized and non-contextualized conditions (Table 7). Critically, the *direction* of the shift was not uniform across models. Gemini, Claude, and Qwen produced systematically higher scores once the learner's country, age, and academic profile were visible, consistent with a leniency or "halo" effect triggered by a plausible learner persona. DeepSeek showed the opposite pattern, becoming stricter once context was supplied. GPT-5.1 was the only model whose scores did not shift significantly, behaving as context-invariant on this corpus.

Model	t-statistic	p-value	Significant ($\alpha = 0.05$)
Gemini 3.5 Flash	5.818	< 0.00001	Yes
DeepSeek V4 Flash	-2.486	0.013	Yes
GPT-5.1	-1.671	0.095	No
Claude Opus 4.8	3.682	< 0.001	Yes
Qwen 3.7 Plus	10.766	< 0.00001	Yes

Table 7: Paired t-test, context vs. no-context total scores (ICNALE).

In contrast, repeating the same experiment on EFCamDat (N = 1,612) as an out-of-sample robustness check revealed that the context effect was significant for *all five* models, and — unlike on ICNALE — all five shifted in the *same* direction. Every model scored contextualized essays higher than their non-contextualized counterparts (Table 8). Gemini showed the largest leniency shift (+1.175), followed by DeepSeek (+1.025) and GPT-5.1 (+0.685), while Claude (+0.266) and Qwen (+0.278) were comparatively the most stable.

Model	Mean Diff (Context − No Context)	t-statistic	p-value	Significant ($\alpha = 0.05$)
Gemini 3.5 Flash	1.175	57.332	< 0.00001	Yes
DeepSeek V4 Flash	1.025	27.047	< 0.00001	Yes
GPT-5.1	0.685	31.755	< 0.00001	Yes
Qwen 3.7 Plus	0.278	14.950	< 0.00001	Yes
Claude Opus 4.8	0.266	14.372	< 0.00001	Yes

Table 8: Paired t-test, context vs. no-context total scores (EFCamDat).

4.2.2 Decomposing Bias: Demographic vs. Structural Factors

To understand what drives these shifts, Table 9 decomposes the ICNALE context effect across seven candidate factors using the *osa*, *osd*, and *csd* metrics. The resulting pattern is consistent across all five models: Age and Grade emerge as the two most influential factors, with *csd* values reaching 0.77–1.27 for every model except Claude under Grade and peaking at 1.844 for DeepSeek under CEFR. Country contributes a moderate but non-negligible effect ($csd \approx 0.43$–0.91), whereas Sex and Topic remain the least influential factors throughout, with *csd* typically below 0.2. DeepSeek and Qwen display the highest overall sensitivity to demographic metadata, each exceeding $csd = 1.0$ on at least two factors, while Claude and GPT-5.1 remain comparatively stable across most dimensions. These results indicate that demographic metadata is not processed as inert background information; rather, attributes such as age and academic grade level appear to trigger measurable adjustments to the underlying scoring behavior.

However, the factor decomposition on EFCamDat (Table 10) shows a much weaker effect compared to ICNALE. In stark contrast to the primary dataset, no *csd* value crosses the critical 0.5 threshold. The conditional shift for the native language (L1) is negligible (0.053–0.125), and the CEFR effect peaks at only 0.355 for Gemini. This indicates that while context causes an overall leniency shift, the models remain largely indifferent to specific L1 and CEFR tags, failing to reproduce the deep conditional biases observed on ICNALE.

4.2.3 Directional Bias: Country and L1-Specific Shifts

While the aggregate metrics in the tables above quantify *how large* the context-driven distortion is per factor, they average away the sign, obscuring whether specific groups are helped or penalized. To recover that directional information, we drill into the Country and L1 dimensions and compute the signed *csd* per subgroup and per model.

Figure 5 shows the signed *csd* heatmap for the total score on ICNALE. Two countries stand out as reproducible outliers across nearly all five models. **Singapore (SIN) is penalized by every model except DeepSeek**, with the context condition lowering its relative score by −0.31

Metric	Model	Sex	Age	Country	Grade	Acad.	CEFR	Topic
	gemini	2.31	2.526	2.31	2.492	2.35	2.314	2.294
	deepseek	4.07	3.84	3.966	3.357	3.793	4.25	4.098
osa	gpt5	2.225	2.197	2.173	2.112	2.118	2.219	2.18
	claude	1.717	1.724	1.691	1.38	1.875	1.716	1.698
	qwen	2.64	2.751	2.654	2.473	2.712	2.573	2.609
	gemini	0.7	1.087	0.948	1.134	0.784	0.609	0.694
	deepseek	0.477	1.168	0.851	1.236	0.907	1.844	0.533
osd	gpt5	0.197	0.869	0.797	0.799	0.464	0.336	0.198
	claude	0.385	0.875	0.599	0.508	0.781	0.816	0.342
	qwen	1.252	1.785	1.362	1.478	1.314	1.259	1.259
	gemini	0.029	0.842	0.752	0.715	0.127	0.591	0.369
	deepseek	0.269	1.268	0.905	1.042	0.791	1.844	0.12
csd	gpt5	0.005	0.865	0.757	0.77	0.443	0.401	0.073
	claude	0.209	0.77	0.431	0.462	0.537	0.645	0.048
	qwen	0.036	1.029	0.634	0.956	0.179	0.629	0.031

Table 9: Overall subgroup absolute deviation (*osa*), overall subgroup difference (*osd*), and conditional score difference (*csd*) of the context effect, by demographic/structural factor (ICNALE).

to −0.39 for GPT-5.1, Gemini, Claude, and Qwen — the single largest negative deviation any country receives from any model in this analysis. **Pakistan (PAK), conversely, is the most consistently favored group**, receiving a positive deviation from four of the five models, peaking at +0.48 for Qwen. Indonesia (IDN) shows a similar, if more model-specific, favorable pattern, driven almost entirely by Gemini (+0.43). DeepSeek is the clear outlier of the group: it does not share the Singapore penalty (in fact scoring SIN essays +0.06 under context) and instead reserves its largest negative deviation for Hong Kong (−0.35) and China (−0.28), a pattern no other model reproduces at comparable magnitude. This confirms that DeepSeek's context sensitivity is real but structured differently from the other four models.

Applying this view to EFCamDat, Figure 6 shows the signed *csd* by L1 for the total score. Despite the low overall L1 bias noted earlier, a directional breakdown reveals slight but clear group-specific shifts. Japanese and Turkish speakers are generally favored by the majority of evaluators. Japanese receives a notable +0.25 boost from GPT-5 and slight +0.11 from Gemini, while Turkish is scored positively by Gemini (+0.21), GPT-5 (+0.19), and Qwen (+0.13). DeepSeek is again the outlier, strongly penalizing Turkish (−0.28) where most other models favor it — consistent with the divergent, model-specific behavior already observed on ICNALE. On the negative side, Arabic speakers face a substantial penalty from Gemini (−0.20), and interestingly, essays tagged with English as the native language receive distinct negative shifts from GPT-5 (−0.19) and DeepSeek (−0.17).

Metric	Model	L1	CEFR
osa	gemini	1.206	1.206
	deepseek	1.407	1.414
	gpt5	0.810	0.809
	claude	0.504	0.505
	qwen	0.500	0.502
osd	gemini	1.180	1.181
	deepseek	1.020	1.030
	gpt5	0.689	0.686
	claude	0.267	0.286
	qwen	0.281	0.281
csd	gemini	0.073	0.355
	deepseek	0.098	0.283
	gpt5	0.125	0.125
	claude	0.053	0.197
	qwen	0.053	0.168

Table 10: Overall subgroup absolute deviation (*osa*), overall subgroup difference (*osd*), and conditional score difference (*csd*) of the context effect, by factor (EFCamDat).

4.2.4 Category-Level Breakdown of Prompt Sensitivity

Breaking this deviation down by analytic category (Figure 7) on ICNALE shows that the Singapore penalty and the Pakistan/Indonesia bonus are not artifacts of a single scoring dimension but recur across most of the five categories. Singapore is penalized most sharply in Content and Vocabulary (Claude: -0.43 and -0.49; GPT-5.1: -0.36 and -0.37; Gemini: -0.45 and -0.39), and still clearly penalized, though less severely, in Organization and Language Use — the only category where the penalty softens noticeably is Mechanics. Pakistan's favorable treatment is strongest in Organization and Mechanics: Qwen awards Pakistani essays a $+0.66$ boost in Organization and $+0.47$ in Mechanics, and DeepSeek shows its single largest positive deviation anywhere in the dataset for Pakistan in Organization ($+0.38$). Indonesia's bonus is almost entirely a Gemini-specific effect, appearing at similar magnitude in every category (Content $+0.54$, Organization $+0.48$, Mechanics $+0.51$), suggesting Gemini associates the Indonesian context tag with a broadly more favorable prior. A notable anomaly is DeepSeek's $+0.59$ Mechanics spike for Japan. As the largest single deviation, it appears to be an isolated prompt-sensitivity artifact rather than a stable bias.

Overall, whether examining total scores, specific demographic tags (Country and L1), or fine-grained category metrics, the cross-corpus analysis reinforces a clear conclusion: LLMs do not maintain objective neutrality when exposed to demographic metadata. Instead of evaluating essays strictly on their textual merit, contextualization systematically distorts the grading process, leading to both the inflation and deflation of scores along specific demographic lines and introducing distinct, model-specific patterns of bias.

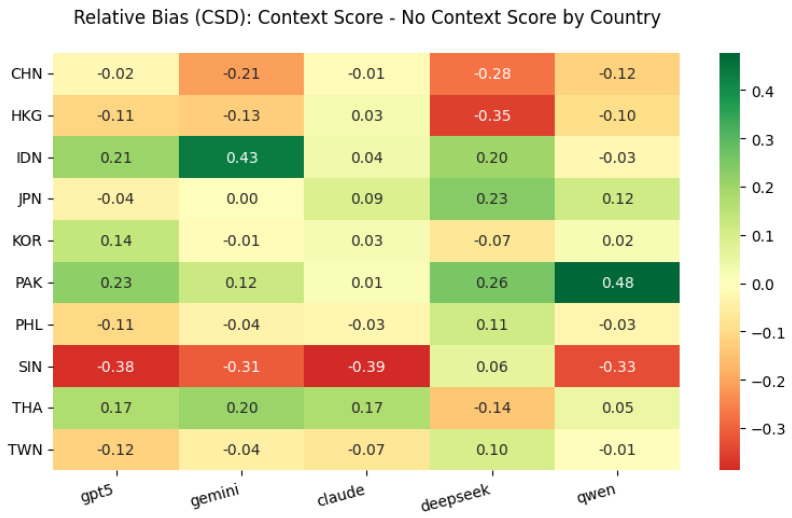


Figure 5: Relative bias (signed *csd*): total-score deviation between contextualized and non-contextualized scoring, by country and model (ICNALE).

4.3 Pure Prompt Sensitivity: Country Injection Experiment

To isolate algorithmic bias from actual text quality, we conducted a counterfactual country injection experiment on the ICNALE dataset. We selected 200 essays in total — exactly 20 essays from each of the 10 represented countries. Every single essay was evaluated 10 times by each model, with the demographic context tag swapped for a different country in each iteration while the underlying text remained perfectly identical. By calculating the internal deviation of each score from the essay's average score across all 10 tags, we eliminated linguistic variables and directly measured the pure prompt sensitivity of the evaluators.

4.3.1 Baseline AI Score Volatility

First, we evaluated the overall score volatility, defined as the average standard deviation of scores awarded to identical essays across the ten injected country tags (Table 11).

Model	Average Volatility (Std Dev)
DeepSeek V4 Flash	0.779
GPT-5.1	0.548
Claude Opus 4.8	0.462
Gemini 3.5 Flash	0.448
Qwen 3.7 Plus	0.448

Table 11: AI score volatility (standard deviation of scores for identical essays across country variations).

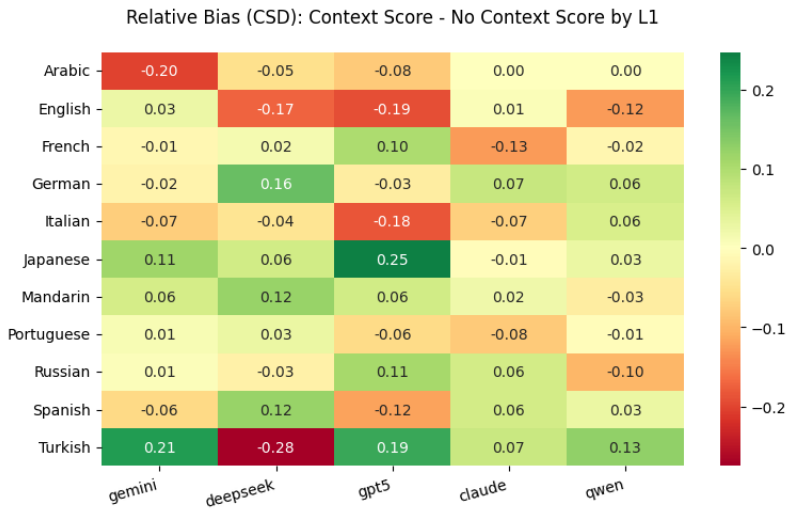


Figure 6: Relative bias (signed *csd*): total-score deviation between contextualized and non-contextualized scoring, by L1 and model (EFCamDat).

DeepSeek demonstrated the highest sensitivity to the prompt context, with an average standard deviation of 0.779 points for the exact same text. GPT-5.1 showed moderate volatility (0.548), while Claude, Gemini, and Qwen proved to be comparatively the most stable, clustering around a standard deviation of 0.448 to 0.462. The presence of non-zero volatility across all evaluated LLMs confirms their inherent susceptibility to prompt-driven variation in essay quality assessment.

4.3.2 Statistical Significance of Injected Bias

To determine whether the directional deviations observed in the internal bias heatmap (Figure 8) were systematic or merely random generative noise, we analyzed the results using a one-sample t-test ($H_0 = 0$) supplemented by a Benjamini-Hochberg False Discovery Rate (FDR) correction and Cohen's d effect size evaluation.

The statistical correction successfully filtered out a significant amount of noise. While the raw heatmap suggests numerous minor deviations, the FDR-corrected results reveal that only Gemini exhibits statistically significant, reproducible bias in its internal scoring logic (Table 12).

Model	Country	Mean Deviation	Raw p-value	Cohen's d	Adj p-value	Significant?	Direction
Gemini	HKG	-0.129	0.0000	-0.293	0.0025	Yes	Penalizes
Gemini	TWN	-0.103	0.0010	-0.236	0.0170	Yes	Penalizes
Gemini	PAK	0.112	0.0002	0.264	0.0062	Yes	Favorable

Table 12: Statistically significant country-specific biases after Benjamini-Hochberg (FDR) correction.

Gemini systematically penalizes essays tagged with Hong Kong (Mean Deviation: -0.129, Adjusted $p = 0.0025$) and Taiwan (Mean Deviation: -0.103, Adjusted $p = 0.0170$). Conversely, the

model shows a statistically significant favorable bias toward essays tagged with Pakistan (Mean Deviation: +0.112, Adjusted $p = 0.0062$). While the Cohen's d values ($|d| \approx 0.23$ to 0.29) indicate small-to-medium effect sizes, the statistical significance proves that Gemini's grading mechanism is reliably altered by these specific regional identifiers. For GPT, Claude, DeepSeek, and Qwen, FDR correction confirmed that all score deviations were merely random noise, showing no systematic bias toward any specific country.

Ultimately, this experiment reveals a critical difference between random scoring instability and true algorithmic bias. While all models exhibit score fluctuations when exposed to different demographic tags, most of this variation is random noise. Gemini, however, demonstrates systematic and reproducible bias against specific regions. This confirms that to ensure fair and objective grading, AI evaluators should assess essays blindly, without access to students' personal backgrounds.

5 Limitations

Despite the rigorous statistical framework applied in this study, several limitations must be acknowledged regarding our methodology, the nature of the evaluated models, and the data utilized.

First, beyond the paired t-tests confirming a global context effect (Section 4.2.1) and the FDR-corrected country injection experiment (Section 4.3), the demographic bias patterns themselves (subgroup metrics (OSA, OSD, CSD), country/L_1 heatmaps, and category-level breakdowns) are all based on the absolute magnitude and visual consistency of observed score shifts, without formal significance testing.

Second, Large Language Models exhibit inherent stochasticity. Because LLMs operate probabilistically, evaluating the exact same essay under identical prompt and context conditions can yield varying scores. While our counterfactual experiments measured targeted volatility across demographic tags, we did not perform a comprehensive, large-scale stability analysis to establish the baseline variance of zero-shot scoring for every essay. Consequently, an underlying margin of random generative instability remains a structural limitation of LLM-based evaluation.

Third, our evaluation framework fundamentally relies on human-assigned grades as the absolute ground truth for calculating accuracy and baseline deviations (such as OSA and OSD). Human raters, however, are rarely perfectly objective and may harbor their own conscious or unconscious demographic biases. By treating human scores as the baseline for neutrality, our metrics may inadvertently measure the models' deviation from human bias, rather than a deviation from true objective fairness.

Fourth, the proprietary nature of the evaluated models presents a fundamental barrier to reproducibility. Accessed primarily via commercial APIs, these models function as what Pasquale (2015) describes as "black boxes" [7]—systems where training data, architectural weights, and alignment mechanisms remain entirely opaque to independent researchers. Furthermore, because corporate providers continuously and silently update model weights, training data, or safety guardrails behind the scenes, the exact replication of the bias patterns observed in this study cannot be guaranteed over time.

Finally, the scale and specificity of our chosen datasets restrict the broad generalizability of our findings. While carefully curated and structured, both the ICNALE ($N = 640$) and EFCamDat ($N = 1,612$) subsets represent relatively small sample sizes within the broader context of global language learning. Additionally, EFCamDat is highly specific to a particular online learning platform's curriculum and task structures. The demographic bias patterns identified in these specific

essay formats may not perfectly extrapolate to other text genres, such as creative writing, academic research papers, or specialized professional communication.

6 Conclusion

As a result of our study, we conducted a detailed analysis of the impact of demographic, linguistic, and structural characteristics on the evaluation behavior of modern large language models.

The length of the text (number of words/tokens) emerged as the primary and most consistent source of bias for all evaluated models. The models demonstrate high sensitivity to essay length, systematically overrating longer texts, regardless of their content quality or relevance to the prompt. Among the evaluators studied, the Claude Opus 4.8 model exhibited the greatest resistance to this effect, maintaining the highest accuracy in alignment with human evaluations with QWK equal to 0.6133 and the lowest tendency to skew scores based on essay length.

Furthermore, models scored with demographic context present showed persistent country-specific biases relative to human scores: ranging from positive bias (for example, for students from Germany or Japan) to systematic under-scoring for other regions (for example, Singapore or Turkey, depending on the model). In contrast, the authors' basic demographic indicators — sex and age — proved to be virtually neutral. For these categories, CSD values remain close to zero, indicating the absence of systematic gender or age bias in the evaluation algorithms.

Beyond these baseline deviations, the mere presence of demographic context in the prompt distorted scores even when the essay itself did not change. On ICNALE, this effect was significant for four of five models, but its direction varied: Gemini, Claude, and Qwen became more lenient with a visible learner persona, DeepSeek became stricter, and GPT-5.1 was context-invariant. On EFCamDat, the effect was significant and positive for all five models. The main drivers also differed by corpus — age and grade on ICNALE, essay length rather than the L1 tag on EFCamDat — and the bias was not uniform: it formed reproducible poles, most notably a Singapore penalty and a Pakistan/Indonesia leniency on ICNALE, recurring across most individual sub-rubrics (CON, ORG, VOC, LNU, MEC).

Finally, the counterfactual country-injection experiment further isolated pure prompt sensitivity from real essay quality. All models showed non-zero score volatility when the same essay was re-tagged with ten different countries, highest for DeepSeek (0.779) and lowest for Claude, Gemini, and Qwen (0.448–0.462). After Benjamini-Hochberg FDR correction, however, only Gemini showed statistically significant, reproducible bias — penalizing Hong Kong- and Taiwan-tagged essays while favoring Pakistan-tagged ones. For the other four models, the deviations were indistinguishable from random noise.

Overall, these results confirm that AI evaluators should assess essays blindly, without access to students' demographic backgrounds, to ensure fair and objective grading.

7 Future Works

Building on the findings and limitations above, several directions could extend this work.

First, most of the bias patterns we report — the subgroup metrics (OSA, OSD, CSD), the country/L1 heatmaps, and the category-level breakdowns — are based on how large and how visually consistent the score shifts look, not on formal statistical testing. Only the country-injection experiment currently goes through the full significance pipeline (one-sample t-tests, Cohen's d, FDR

correction). Applying that same pipeline to the rest of the results would let us say more confidently which patterns are real bias and which are just noise.

Second, LLMs are stochastic: scoring the same essay with the exact same prompt and context twice can still give different scores. We didn't measure this directly — our volatility numbers only compare scores across different country tags, not repeated runs with the same tag. Future work should quantify this baseline noise directly, for example by scoring each essay several times under a fixed prompt and averaging the results, so that demographic effects can be separated more cleanly from ordinary model randomness.

Third, the country-injection experiment in Section 4.3 used a stratified subsample of 200 essays; running it on the full ICNALE corpus, and ideally extending it to other demographic tags like age or gender, would give more statistical power and a more fine-grained bias map. Beyond these three priorities, a few other directions are worth exploring: testing mitigation strategies such as blind grading, per-subgroup score calibration, or explicit fairness instructions in the prompt, to see if they reduce CSD without hurting agreement with human scores (QWK); re-running the same protocol on the same models at a later point in time, since providers update their models and current bias patterns may not be stable; and digging into why bias isn't uniform across the five sub-rubrics (CON/ORG/VOC/LNU/MEC) — for example, by running targeted prompt ablations that isolate specific cues — the nationality label, the implied L1, or the CEFR expectation — one at a time — to better understand effects like the Singapore Vocabulary penalty or the Pakistan Organization/Mechanics bonus we found.

8 Author Contributions

Oksana Yefimova coordinated task allocation among team members, downloaded the ICNALE dataset; requested access to the EFCamDat dataset and finalized its preprocessing; executed the contextualized and non-contextualized prompt evaluation for the majority of models on ICNALE and the contextualized evaluation on EFCamDat; applied the QWK, paired t-test, and subgroup bias metrics (OSA, OSD, CSD) across the experiments; authored the Abstract, Sections 1, 2, 3.2 and 4.1, co-authored the Conclusion, and contributed to the preparation of presentation materials.

Tetiana Mazko downloaded the EFCamDat dataset and carried out its initial preprocessing; evaluated some models under contextualized and non-contextualized prompting on ICNALE and non-contextualized prompting on EFCamDat; designed and executed the counterfactual country injection experiment, applied the statistical framework (one-sample t-tests, Cohen's d, FDR correction) used to isolate pure prompt sensitivity, and generated the analytical heatmaps visualizing CSD and regional bias patterns; authored Sections 4.2, 4.3, 5, and 8, co-authored the Conclusion, and contributed to the preparation of presentation materials.

Liliia Yaroshenko authored Sections 3.1 and 7 of the paper, developed the majority of the presentations and progress reports for all project checkpoints, assisted in the development of the final presentation, and actively participated in discussions of interim results.

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Figure 7: Relative bias (signed *csd*) between contextualized and non-contextualized scoring, by country, model, and analytic category — Content (CON), Organization (ORG), Vocabulary (VOC), Language Use (LNU), and Mechanics (MEC) (ICNALE).

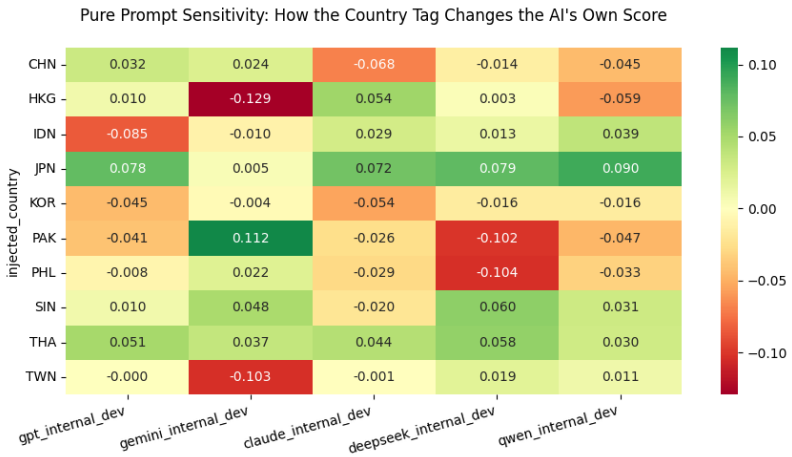


Figure 8: Pure Prompt Sensitivity: Internal score deviation driven solely by the injected country tag.